

First course handout (FCH) of MTH 314A: Multivariate Analysis

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<https://home.iitk.ac.in/~shalab/course4.htm>

1. **Objectives:** The course aims to develop concepts and tools related to multivariate analysis from a statistical perspective.

2. **Prerequisites:** Courses on MTH211A (Theory of Statistics) or MTH418A (Inference 1).

3. **Course Contents:**

Basic properties of random vector: CDF and PDF of random vectors - Moments - Characteristics functions Orthogonal and Polar transformations Generalization of univariate distribution (Multinomial, Dirichlet).

Normal Distribution Theory: Normal data matrix (NDM): characterization and properties Linear forms - Transformation of NDMs, Wishart Distribution, The Hotelling's T^2 Distribution

Estimation and Testing: Maximum likelihood estimation Likelihood ratio test - Union intersection test Simultaneous confidence intervals.

Multivariate Analysis of Variance (MANOVA): Formulation of multivariate one-way classification Likelihood ratio principle

Principal Component Analysis: Principal components - Sampling properties of principal components - Principal component projections.

Factor Analysis: The factor model - Principal factor analysis - Maximum likelihood factor analysis - Goodness of fit - rotation of factors - factor scores

Canonical correlation analysis: Population and sample canonical correlation vectors, variables and coefficients and their properties.

Discrimination Analysis: Fisher's LDA - QDA - Probabilities of misclassification

Cluster Analysis: Distances and similarities - Hierarchical methods - K-means method

Books to be followed:

Anderson, T. W. (2003). An Introduction to Multivariate Statistical Analysis. United Kingdom: Wiley.

Johnson, R. A., Wichern, D. W. (2019). Applied Multivariate Statistical Analysis. United Kingdom: Pearson

Other References:

Mardia, K. V., Bibby, J. M., Kent, J. T. (1979). Multivariate Analysis. United Kingdom: Academic Press.

Muirhead, R. J. (2009). Aspects of Multivariate Statistical Theory. Germany: Wiley.

Hastie, T., Friedman, J., Tibshirani, R. (2009). The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Germany: Springer New York.

Brenner, D., Bilodeau, M. (1999). Theory of Multivariate Statistics. Germany: Springer.

4. Lecture, Tutorial & Lab Schedule & Venue, use of LMS (if any): 3 Lectures per week and one Lab
5. Office Hours: Anytime as per the mutual convenience
6. Evaluation Components & Policies: Exams, Quizzes, Assignments.
7. Course Policies: Attendance, Honesty Practices, Withdrawal (within the limits of DOAA Guidelines) and all others as expected from good students.